

An Introduction to the Go Programming Language



Here's how Google describes Go in its blog post: "Go attempts to combine the development speed of working in a dynamic language like Python with the performance and safety of a compiled language like C or C++. In our experiments with Go, to date, typical builds feel instantaneous; even large binaries compile in just a few seconds." Let's take a quick look at this powerful language.

Go is an open source concurrent programming language published by Google. It is similar to the C programming language, but it has several features that make it unique.

- Compilation of the project is so fast, it feels as if Go is an interpreted language rather than a compiled one. Programs are also easily edited or run on-the-fly.
- It has garbage-collection incorporated into it. So developers do not have to perform memory management as Go takes care of most of this work.
- It has built-in concurrency. So it can support concurrent work that requires permitting both communication and synchronisation.
- Go is a robust and statically-typed language. So several possible bugs can be detected during compilation, and resolved.
- Go has a fully working Web server as part of its standard library.
- There is no need to mess around with *build* configurations or *make* files as Go's built-in build system is very simple.

Development environment set-up

To set up the local development environment, we need the

following pre-requisites on our computer:

- **Text editor:** This is used to store our programs written in the Go language. Examples of a few editors are Windows Notepad, Vim or Vi. The files containing program source code are called source files. They must contain the extension *.go*.
- **Go compiler:** The source code written in files needs to be converted to a language that the CPU understands from the human readable source program. The compiler will compile the source code and convert the code into machine language, which the CPU can run as per program instructions.
 - Download the Go compiler from <https://golang.org/dl/>
 - The GOPATH environment variable set-up specifies the location of the current Go workspace. It is the only environment variable we need for the Go code.

The workspace can be located anywhere one wants. In my case I have used *\$HOME/dev*.

```
$ mkdir $HOME/dev
$ export GOPATH=$HOME/dev
```

To test if everything works correctly, type the following